

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407119
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Stecewycz; Joseph

Automotive battery with integral electrical contact slots

Abstract

An automotive battery includes a positive contact slot in a battery housing, the slot opening into a positive slot encasement inside the battery for insertion of a positive electrical prong for electrical connection to a contact region on a positive battery post; and, a negative contact slot in the battery housing, the slot opening into a negative slot encasement inside the battery for insertion of a negative electrical prong for electrical connection to a contact region on a negative battery post.

Inventors:	Stecewycz; Joseph (Groton, MA)
Applicant:	Stecewycz; Joseph (Groton, MA)
Family ID:	82800629
Assignee:	Cythorean Mandelbrot, LLC (Groton, MA)
Appl. No.:	17/734107
Filed:	May 01, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220263257 A1	Aug. 18, 2022

Related U.S. Application Data

us-provisional-application US 63194135 20210527

Publication Classification

Int. Cl.: H02J7/00 (20060101); H01M10/44 (20060101); H01M50/503 (20210101);
H01M50/543 (20210101); H01R11/28 (20060101); H02J7/14 (20060101); H02J7/34
(20060101)

U.S. Cl.:

CPC **H01R11/289** (20130101); **H01M10/44** (20130101); **H01M50/503** (20210101);
H01M50/543 (20210101); **H01R11/288** (20130101); **H02J7/342** (20200101);
H01M2220/20 (20130101)

Field of Classification Search

CPC: H01R (11/287); H01R (11/289); H01R (11/288); H01R (11/00); H01R (11/18); H02J
(7/342); H01M (50/543); H01M (50/503); H01M (50/531); H01M (10/44); H01M
(10/06); H01M (10/12); H01M (10/10); H01M (2220/20)

USPC: 320/103; 320/132

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3466453	12/1968	Greenberg	320/DIG.15	B60D 1/62
3625774	12/1970	Rodgers	439/500	H01R 11/287
4224383	12/1979	Taylor	429/7	H01M 50/289
4424264	12/1983	McGuire	429/179	H01M 50/147
4637965	12/1986	Davis	429/178	H01R 11/287
4807895	12/1988	Thomas	180/68.5	H02G 3/00
4885524	12/1988	Wilburn	320/DIG.15	H01M 50/55
5146149	12/1991	Nilssen	320/112	H01M 10/46
5236792	12/1992	Larkin	429/121	H01M 50/278
5804770	12/1997	Tanaka	174/138F	H01M 50/591
5921809	12/1998	Fink	439/679	H01R 11/289
6147471	12/1999	Hunter	320/105	H01R 11/287
6343949	12/2001	Floyd	439/284	H01R 13/642
9368912	12/2015	Sullivan	N/A	H01R 11/288
2001/0012581	12/2000	Zhang	N/A	N/A
2019/0199017	12/2018	Stecewycz	N/A	H02J 7/0034

Primary Examiner: Pacheco; Alexis B

Background/Summary

FIELD OF THE INVENTION

(1) This invention relates to automotive batteries and, in particular, to automotive batteries incorporating integral electrical contact slots to facilitate a battery charging operation without the need for a conventional battery booster cable with battery terminal clamps.

BACKGROUND OF THE INVENTION

(2) FIG. 1 is a simplified diagrammatical illustration of a typical, conventional automotive lead acid battery **10** that includes a positive battery terminal **12**, a negative battery terminal **14**, a battery casing **16**, and a battery cover **18**. The battery **10** includes battery terminal bases **22** at the battery

terminals **12**, **14**. The battery cover **18** is sealed to the battery casing **16**, forming a leakproof battery housing **20**, to prevent leakage of internal battery electrolyte solution from the battery **10**. Vent caps **24** are provided in the battery cover **18** to prevent buildup of gases in the battery housing **20**.

(3) As shown in FIG. 2, a conventional booster cable **30** typically includes “positive” battery terminal clamps **32** and “negative” battery terminal clamps **34** featuring spring-loaded serrated jaws for attachment. The positive battery terminal clamps **32** are connected to opposite ends of a positive insulated electrical conductor **36** and the negative battery terminal clamps **34** are connected to opposite ends of a negative insulated electrical conductor **38**. The insulated electrical conductors **36**, **38** each have sufficient cross-sectional areas to safely conduct high amperage engine starter current.

(4) FIG. 3 is a simplified, generic diagrammatical illustration of the battery **10** with the battery cover **18** and the front and sides of the battery casing **16** removed to show typical internal battery components. The positive battery terminal **12** is electrically connected to a positive battery post **42** and the negative battery terminal **14** is electrically connected to a negative battery post **44**, where both the battery posts **42**, **44** extend below the battery cover **18** into the battery casing **16**.

(5) The battery **10** includes a plurality of battery cells **52** arranged in a parallel configuration with a plurality of intercell partitions **54** separating adjacent battery cells **52**. The number of battery cells **52** determines the total voltage output of the battery **10**. In the configuration shown, each battery cell **52** includes two positive plates **62** and two negative plates **64**, where each positive plate **62** is separated from an adjacent negative plate **64** by a plate separator **66**. The plates **62**, **64** and the plate separators **66** are normally immersed in battery electrolyte solution (not shown).

(6) The six battery cells **52** are electrically connected in series by five intercell connectors **56**. Each intercell connector **56** electrically connects two positive plates **62** in one battery cell **52** to two negative plates **64** in an adjacent battery cell **52**. The positive battery post **42** is mounted on a positive plate strap **58** that is electrically connected to at least two positive plates **62**. Similarly, the negative battery post **44** is mounted on a negative plate strap **68** that is electrically connected to at least two negative plates **64**. Electrical connection to the battery **10** is conventionally made at the respective outer surfaces of the positive battery terminal **12** and the negative battery terminal **14**.

(7) As is tolerated in common usage, the booster cable **30** suffers from the shortcoming that the serrated jaws of the booster cable clamps **32**, **34**, typically made from copper or a copper alloy, produce indentations and scratches on the softer lead alloy battery terminals **12**, **14**, or on lead alloy engine battery terminal clamps (not shown) secured to the battery terminals **12**, **14**. Over time, clamping and unclamping operations produces deterioration of the lead alloy surfaces from repeated attachment of the booster cable **30**, especially when a user twists and turns the booster cable clamps **32**, **34** to insure good electrical connections.

(8) Another shortcoming is that one or both of the booster cable clamps **32**, **34** may accidentally make contact with the vehicle chassis or metal engine components when a user attaches or removes the conventional booster cable **30**. A further shortcoming is that the process of connecting the booster cable **30** typically requires the use of both hands to prevent the booster cable clamps **32**, **34** from accidentally contacting each other. Additionally, there is a finite probability that an electrical spark created between a booster cable clamp and a battery terminal post may ignite hydrogen gas produced by battery overcharging.

(9) What is needed is a battery charging apparatus that overcomes the limitations of the prior art by: (i) providing a method of electrically connecting automotive batteries without causing surface damage to battery terminals and to battery clamps, (ii) assuring proper polarity when connecting a source battery to a discharged battery, and (iii) preventing generation of electrical sparks that may cause hydrogen gas ignition.

BRIEF SUMMARY OF THE INVENTION

(10) The invention results from the observation that an electrical connection to a battery for the

purpose of charging can be made via encased electrical contact slots, inside the battery housing, terminating at battery posts, thus bypassing the external battery terminals.

(11) In an aspect of the present invention, an automotive battery with a positive battery post providing an electrical path between a positive battery terminal and a positive plate strap, and a negative battery post providing an electrical path between a negative battery terminal and a negative plate strap includes: a positive contact slot encasement inside the battery, the positive contact slot encasement enclosing a positive contact slot and at least a portion of the positive battery post, the positive contact slot having a first opening in a battery housing, the positive contact slot providing access to a positive battery post contact region so as to allow insertion of a positive electrical prong into the first opening whereby a positive prong contact surface is placed into electrical contact with the positive battery post; and, a negative contact slot encasement inside the battery, the negative contact slot encasement enclosing a negative contact slot and at least a portion of the negative battery post, the negative contact slot having a second opening in the battery housing, the negative contact slot providing access to a negative battery post contact region so as to allow for insertion of a negative electrical prong into the second opening whereby a negative prong contact surface is placed into electrical contact with the negative battery post.

(12) In another aspect of the present invention, an automotive battery with (i) a positive battery terminal on a battery housing connected to a positive battery post inside the battery housing, the positive battery post providing an electrical path between the positive battery terminal and a positive plate strap in the battery housing, and (ii) a negative battery terminal on the battery housing connected to a negative battery post inside the battery housing, the negative battery post providing an electrical path between the negative battery terminal and a negative plate strap in the battery housing, includes: a positive contact slot encasement inside the battery housing, the positive contact slot encasement having a positive cavity side wall attached to the battery housing, a positive cavity end wall attached to the battery housing, and a positive cavity base attached to both positive cavity walls, the positive contact slot encasement enclosing at least a portion of the positive battery post; a positive contact slot having a first opening in the battery housing to provide for insertion of a positive electrical prong into the positive contact slot encasement, the positive electrical prong having a contact surface for electrically contacting a positive post electrical contact region on a positive battery post surface when inserted into the positive contact slot encasement; a negative contact slot encasement inside the battery housing, the negative contact slot encasement having a negative cavity side wall attached to the battery housing, a negative cavity end wall attached to the battery housing, and a negative cavity base attached to both negative cavity walls, the negative contact slot encasement enclosing at least a portion of the negative battery post; and a negative contact slot having a second opening in the battery housing to provide for insertion of a negative electrical prong into the negative contact slot encasement, the negative electrical prong having a contact surface for electrically contacting a negative post electrical contact region on a negative battery post surface when inserted into the negative contact slot encasement.

(13) In yet another aspect of the present invention, a charging system suitable for charging a drained battery using a source battery includes: a source battery connection adapter having a positive source electrical plug for electrically connecting to a source positive battery post via a source positive contact slot, and a negative source electrical plug for electrically connecting to a source negative battery post via a source negative contact slot; a drained battery connection adapter having a positive insulated electrical conductor for electrically connecting to a drained positive battery terminal, and a negative insulated electrical conductor for electrically connecting to a drained negative battery terminal; and, an adapter connection cable configured to be electrically connected both to the source battery connection adapter and to the drained battery connection adapter so as to form a battery interconnection cable assembly such that, when the battery interconnection cable assembly is electrically connected both to the source battery and to the drained battery, the battery interconnection cable assembly provides a positive electrical path

between the source positive battery post and the drained positive battery terminal, and provides a negative electrical path between the source negative battery post and the drained negative battery terminal.

(14) The additional features and advantage of the disclosed invention is set forth in the detailed description which follows, and will be apparent to those skilled in the art from the description or recognized by practicing the invention as described, together with the claims and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The uses and advantages of the present invention will be more fully appreciated as the same becomes better understood from the following detailed description of the present invention when viewed in conjunction with the accompanying figures, in which:

(2) FIG. 1 is a diagrammatical illustration of an automotive lead acid battery showing a battery housing with positive and negative battery terminals on a battery cover, in accordance with the present state of the art;

(3) FIG. 2 is a diagrammatical illustration of a conventional booster cable including booster cable clamps connected to opposite ends of a pair of insulated electrical conductors, in accordance with the present state of the art;

(4) FIG. 3 is a diagrammatical illustration of the internal configuration of the battery of FIG. 1 with the battery cover removed to show battery posts, internal battery plates, and intercell connectors;

(5) FIG. 4 is a simplified diagrammatical illustration of an automotive battery with a positive contact slot and a negative contact slot at the front of a battery cover, in accordance with the present invention;

(6) FIG. 5 is an assembly cutaway view of part of the battery of FIG. 4 showing a positive battery post, the positive contact slot, and a positive contact slot encasement;

(7) FIG. 6 is a cross-sectional plan view of the battery of FIG. 4 taken across a section plane A-A as shown in FIG. 4;

(8) FIG. 7 is a cross-sectional detail view of the positive battery terminal of FIG. 4 and the positive contact slot, taken across a section plane B-B as shown in FIG. 4;

(9) FIG. 8 is a simplified diagrammatical illustration of an automotive battery with positive and negative contact slots at the sides of a battery cover, in accordance with the present invention;

(10) FIG. 9 is a cross-sectional plan view of the battery of FIG. 8 taken across a section plane C-C as shown in FIG. 8;

(11) FIG. 10 is a simplified diagrammatical illustration of an automotive battery with a vertical positive contact slot and a negative contact slot at the front of a battery housing, in accordance with the present invention;

(12) FIG. 11 is a cross-sectional detail cutaway view of the battery of FIG. 10 showing a positive battery post, the positive contact slot, and a positive contact slot encasement;

(13) FIG. 12 is a cross-sectional plan view of the battery of FIG. 10 taken across a section plane D-D as shown in FIG. 10;

(14) FIG. 13 is a cross-sectional plan view of the battery of FIG. 10 taken across a section plane E-E as shown in FIG. 10;

(15) FIG. 14 is a cross-sectional detail cutaway view of the battery of FIG. 10 showing the positive battery post, an alternate configuration of a positive contact slot, and an alternate configuration of a positive contact slot encasement;

(16) FIG. 15 is a cross-sectional detail cutaway view of the battery of FIG. 10 showing the positive battery post, another configuration of a positive contact slot, and another configuration of a positive

contact slot encasement;

- (17) FIG. **16** is a simplified diagrammatical illustration of an automotive battery with a vertical positive contact slot at the left side of a battery housing, a vertical positive contact slot at the front of the battery housing, a vertical negative contact slot at the right side of the battery housing, and a negative contact slot at the front of the battery housing, in accordance with the present invention;
- (18) FIG. **17** is a cross-sectional detail cutaway view of the battery of FIG. **16** showing a positive battery terminal, the two positive contact slots, and a positive contact slot encasement;
- (19) FIG. **18** is an isometric diagrammatical cutaway view of the positive battery terminal of FIG. **4** showing the configuration of the positive contact slot and a positive electrical plug positioned for insertion into the positive contact slot;
- (20) FIG. **19** is an isometric diagrammatical cutaway view of the battery terminals of FIG. **4** showing alternative positive and negative contact slot configurations, and a positive electrical plug positioned for insertion into the positive contact slot;
- (21) FIG. **20** is a diagrammatical view of alternative polarized positive and negative contact slots with perimeter ridges in a battery, in accordance with the present invention;
- (22) FIG. **21** is a diagrammatical view of alternative polarized positive and negative contact slots of different sizes in a battery, in accordance with the present invention;
- (23) FIG. **22** is a diagrammatical view of alternative offset jog rectangular polarized positive and negative contact slots in a battery, in accordance with the present invention;
- (24) FIG. **23** is a simplified diagrammatical illustration of a contact slot encasement configuration for use in the automotive batteries of FIGS. **4** and **8**;
- (25) FIG. **24** is an isometric diagrammatical illustration of a battery post showing a lateral convex electrical contact region on the surface;
- (26) FIG. **25** is a simplified diagrammatical illustration of an alternative contact slot encasement for use in the battery of FIG. **16**;
- (27) FIG. **26** is an isometric diagrammatical illustration of a battery post showing a longitudinal convex electrical contact region on the surface;
- (28) FIG. **27** is a simplified diagrammatical illustration of a battery with a positive contact slot and a negative contact slot in the top surface of a battery cover;
- (29) FIG. **28** is a cross-sectional view F-F of a positive battery post and the positive contact slot of the battery in FIG. **27**, showing a positive electrical plug positioned for insertion into the positive contact slot;
- (30) FIG. **29** is a detail isometric view of the positive electrical plug of FIG. **28** showing a substantially parallelepiped electrical prong with side edges, top and bottom rectangular surfaces, and a leading concave contact edge;
- (31) FIG. **30** is a simplified diagrammatical illustration of a battery with a positive contact slot and a negative contact slot in a perimeter front edge of a battery cover, in accordance with the present invention;
- (32) FIG. **31** is a simplified diagrammatical illustration of a battery with a positive contact slot in a perimeter left side edge and a negative contact slot in a perimeter right side edge of a battery cover, in accordance with the present invention;
- (33) FIG. **32** is a simplified diagrammatical illustration of a battery with a positive contact slot in a left side edge and a negative contact slot in a right side edge of a battery casing, in accordance with the present invention;
- (34) FIG. **33** is a simplified diagrammatical illustration of a battery with positive and negative contact slots at left and right front corners of a battery cover, in accordance with the present invention;
- (35) FIG. **34** is a cross-sectional plan view of the battery of FIG. **6** showing alternative contact slot encasements;
- (36) FIG. **35** is a cross-sectional plan view of the battery of FIG. **9** showing alternative contact slot

encasements;

(37) FIG. **36** is a cross-sectional plan view of the battery of FIG. **12** showing alternative contact slot encasements;

(38) FIG. **37** is a simplified diagrammatical illustration of a side terminal automotive battery with a vertical positive contact slot and a vertical negative contact slot in a battery casing, and with a positive contact slot and a negative contact slot in a battery cover, in accordance with the present invention;

(39) FIG. **38** is a cross-sectional detail cutaway view of the battery of FIG. **34** showing a positive battery post and two positive electrical plugs positioned for insertion into the two positive contact slots;

(40) FIG. **39** is an isometric diagrammatical illustration of the battery post of FIG. **34** showing a lateral convex electrical contact region on the surface;

(41) FIG. **40** is a simplified diagrammatical illustration of a side terminal automotive battery with a horizontal positive contact slot and a horizontal negative contact slot in a battery casing, and with a positive contact slot and a negative contact slot in a battery cover, in accordance with the present invention;

(42) FIG. **41** is a cross-sectional detail cutaway view of the battery of FIG. **37** showing a positive battery post and two positive electrical plugs positioned for making electrical connections to the positive battery post;

(43) FIG. **42** is an isometric diagrammatical illustration of the battery post of FIG. **38** showing a longitudinal convex electrical contact region on the surface;

(44) FIG. **43** is a simplified diagrammatical illustration of a side-terminal battery with a positive contact slot and a negative contact slot in the top surface of a battery cover, in accordance with the present invention;

(45) FIG. **44** is an isometric diagrammatical illustration of an electrical prong positioned to make electrical contact with a positive battery post in the battery of FIG. **40**;

(46) FIG. **45** is an isometric diagrammatical illustration of a slot encasement configured for use in the automotive battery of FIG. **40**;

(47) FIG. **46** is an isometric diagrammatical illustration of an alternative slot encasement configured for use in the automotive battery of FIG. **40**;

(48) FIG. **47** is a simplified diagrammatical illustration of a side-terminal battery with a positive contact slot in a left side of a battery casing and a negative contact slot in a right side of the battery casing, in accordance with the present invention;

(49) FIG. **48** is an isometric diagrammatical illustration of an electrical prong positioned to make electrical contact with a positive battery post in the automotive battery of FIG. **44**;

(50) FIG. **49** is an isometric diagrammatical illustration of a contact slot encasement configured for use in the automotive battery of FIG. **44**;

(51) FIG. **50** is an isometric diagrammatical illustration of a setup for charging a battery, in which a source battery and a drained battery have the same contact slot configurations, showing a source battery connection adapter, a drained battery connection adapter, and an adapter connection cable, in accordance with the present invention;

(52) FIG. **51** is a detail view of the source battery connection adapter of FIG. **47**;

(53) FIG. **52** is a detail view of the adapter connection cable of FIG. **47**;

(54) FIG. **53** a detail view of a battery adapter cable that can be used in place of the source battery connection adapter and the adapter connection cable of FIG. **47**;

(55) FIG. **54** a detail view of a battery interconnection cable that can be used in place of both battery connection adapters and the adapter connection cable of FIG. **47**;

(56) FIG. **55** is a detail view of a battery connection adapter having a positive electrical socket with a perimeter groove, in accordance with the present invention;

(57) FIG. **56** is a detail view of a battery connection adapter with a positive electrical socket

oriented at an angle to a negative electrical socket, in accordance with the present invention;

(58) FIG. 57 is a detail view of a battery connection adapter with a round positive electrical socket and a round negative electrical socket, in accordance with the present invention;

(59) FIG. 58 is a detail view of a coaxial battery connection adapter including a central conductive positive sleeve and an outer conductive negative cylindrical shell, in accordance with the present invention;

(60) FIG. 59 is an isometric diagrammatical illustration of a setup for charging a battery, in which a source battery and a drained battery have different contact slot configurations, in accordance with the present invention;

(61) FIG. 60 is an isometric diagrammatical illustration of a setup for charging a battery, in which a drained battery includes a battery side mount clamp, in accordance with the present invention;

(62) FIG. 61 is an isometric diagrammatical illustration of a setup for charging a battery, in which a drained battery has no contact slots, showing a battery connection adapter with a polarized battery clamp, in accordance with the present invention; and

(63) FIG. 62 is an alternative diagrammatical illustration of setup for charging a battery, in which the drained battery has no contact slots, showing a battery connection adapter with a booster cable clamp adapter, in accordance with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(64) Various exemplary embodiments of the present invention are described in the specification below, each with reference to the appropriate Figure(s). It should be understood that, for clarity of illustration, not all disclosed battery and cable components are shown to the same scale, or in correct proportion to one another, and should not be taken as literal illustrations of actual battery and cable components. In addition, although some battery components and innovative modifications are presented with straight edges, angular corners, and flat surfaces, present-day manufacturing methods can produce these components having, rounded edges, corner fillets, and curved surfaces. Also, the S.A.E. top-terminal batteries and side-terminal batteries shown are merely representative of the various configurations of automobile lead acid batteries in widespread usage, and the illustrations shown are generalized and should not be taken as limiting the application of the present invention. In particular, the disclosed battery charging cable components can be utilized for L-terminal batteries, for stud-terminal batteries, and for combined S.A.E. and stud type batteries.

(65) The present invention provides novel battery housing configurations which enable an electrical charging connection to be made internally, at battery posts, instead of externally, at battery terminals. These novel configurations provide for two or more polarized recessed electrical contact slots, extending from the battery cover and/or the battery casing, into encasements in the battery housing, as described in greater detail below. The electrical contact slots: (i) enable electrically connecting to a battery without causing surface damage to battery terminals or to engine battery clamps, (ii) assure proper polarity between a source battery and a discharged battery, (iii) reduce the possibility of hydrogen gas ignition from electrical sparks at a battery terminal, and (iv) provide for top, front, and side electrical charging connections on the battery to accommodate different engine compartment layouts.

(66) An exemplary embodiment of a battery **100**, configured in accordance with the present invention, is shown in the simplified diagrammatical illustration of FIG. 4. The battery **100** includes a positive battery terminal **102**, a negative battery terminal **104**, a battery cover **106**, and a battery casing **110**. The battery cover **106** is typically permanently attached to the battery casing **110** by a battery cover flange seal **108** using thermal bonding or chemical adhesive means to form a battery housing **109**. The battery **100** also includes a positive contact slot **112** and a negative contact slot **114**, both contact slots **112**, **114** passing through the battery cover flange seal **108** and the battery casing **110** and into the battery **100**. The positive contact slot **112** and the negative contact slot **114** mate with external electrical connectors (not shown) for charging purposes, as

described in greater detail below.

(67) FIG. 5 is a simplified detail diagrammatical illustration of part of the battery **100** at the positive battery terminal **102**. For clarity of illustration, the wall thicknesses of the battery cover flange seal **108** and the battery casing **110** have been exaggerated to allow cross-sectional hatch lines to be seen. The assembly cutaway view shows a positive plate strap **124** electrically connected to an internal positive battery plate **126**. The positive battery terminal **102** is part of a positive battery post **122**, where the positive battery post **122** lies entirely inside the battery **100**.

(68) A positive contact slot encasement **130** encloses the positive contact slot **112** and part of the positive battery post **122**. The positive contact slot encasement **130** is configured to provide a leak-proof seal at the positive battery post **122** so as to prevent battery electrolyte solution (not shown) from seeping into the positive contact slot **112**. In an exemplary embodiment, the positive contact slot encasement **130** may be formed as an integral part of the battery cover **106**. Alternatively, the positive contact slot encasement **130** may be fabricated as a separate component, and bonded to a battery cover internal surface **107** during battery assembly, as indicated by a bonding interface **138**. An adhesive or sealant (not shown) may be applied between: (i) a positive encasement side surface **134** and a battery casing internal surface **118**, and (ii) a positive encasement front surface **136** and the battery casing internal surface **118** to further prevent leakage of the battery electrolyte solution.

(69) In an exemplary embodiment, the positive contact slot **112** provides openings in the battery cover flange seal **108** and the battery casing **110**, and extends into the positive contact slot encasement **130**, whereby a positive post electrical contact region **128** on the convex surface of the positive battery post **122** is made accessible to an external electrical connector (not shown). It can be appreciated that the electrical contact surface area provided by the positive post electrical contact region **128** can exceed the aggregate electrical contact surface area available on a battery terminal engaged by the opposed teeth of a conventional jumper cable battery clamp. Providing this larger electrical interface area, i.e., the positive post electrical contact region **128**, can advantageously result in a smaller voltage drop at the positive battery post **122** in comparison to using a conventional jumper cable clamp at the positive battery terminal **102**.

(70) FIG. 6 is a cross-sectional plan view A-A of the battery **100**, taken across the section plane A-A as indicated in FIG. 4. In this view, it can be seen that the positive post electrical contact region **128** is directly accessible to a suitably configured electrical prong (not shown) that is sized and shaped to enter the positive contact slot **112**, to pass through the battery cover flange seal **108**, and to pass through the battery casing **110** into the positive contact slot encasement **130**. The illustration more clearly shows the positive encasement side surface **134** and the positive encasement front surface **136** as disposed against the battery casing internal surface **118**. A negative battery post **142** is partially enclosed by a negative contact slot encasement **140** that also encloses the negative contact slot **114**. Similarly, a negative post electrical contact region **148** on the negative battery post **142** is directly accessible to a negative electrical prong (not shown) via the negative contact slot **114**. Note that a negative encasement front surface **146** and a negative encasement side surface **144** are disposed against the battery casing internal surface **118** to prevent seepage of battery electrolyte solution (not shown) into the negative contact slot **114**.

(71) FIG. 7 is a cross-sectional detail view B-B of the positive battery terminal **102** and the positive contact slot **112** of the battery **100**, taken across the section plane B-B through the battery cover **106**, as indicated in FIG. 4 except that, for clarity of illustration, the section plane B-B does not pass through the positive battery terminal **102**, nor through the positive battery post **122**. The positive battery post **122** functions as an electrical conductor from the positive battery terminal **102**, to the positive plate strap **124**, and to the positive battery plate **126**. As can be appreciated by one skilled in the relevant art, the battery **100** can be charged, or can be used to provide an electrical charge to another battery, by inserting a compatible electrical plug (not shown) into the positive contact slot **112** to make electrical contact with the positive post electrical contact region **128** on the positive battery post **122**.

(72) The illustration provides additional detail showing that the positive contact slot **112** extends through the battery cover flange seal **108**, through the battery casing **110**, and into the positive contact slot encasement **130** to expose the positive post electrical contact region **128**. A positive encasement through hole **152** in the positive contact slot encasement **130** may have substantially the same diameter as the positive battery post **122** at a circular leakproof interface **154**, so as to prevent battery electrolyte solution (not shown) from seeping into the positive contact slot **112**. The positive encasement front surface **136** is disposed against the battery casing internal surface **118** to provide for optional application of an electrolyte-resistant sealant **156** during a battery fabrication step. The battery cover flange seal **108** is typically permanently bonded to the battery casing **110**, as represented by a bonding interface **158**.

(73) FIG. **8** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **160**, in accordance with the present invention, that includes a positive battery terminal **162** and a negative battery terminal **164** on a battery cover **166** attached to a battery casing **170** with a battery cover flange seal **168**. The battery cover **166** and the battery casing **170** form a battery housing **169**. In the configuration shown, a positive contact slot **172** provides an opening in a left flange **176** and in a left casing wall **182**. A negative contact slot **174** (hidden) provides an opening in a right flange **178** and a right casing wall **184** (both hidden). The positive contact slot **172** includes an alignment groove **173**. The negative contact slot **174** is a mirror image of the positive contact slot **172**, and together form a polarized electrical contact slot pair.

(74) FIG. **9** is a cross-sectional plan view C-C of the battery **160**, taken across the section plane C-C as indicated in FIG. **8**. This view more clearly shows a positive battery post **186** and the positive contact slot **172** enclosed by a positive contact slot encasement **180**. Also seen is the left flange **176** disposed against the left casing wall **182**, and the right flange **178** disposed against the right casing wall **184**. A positive post electrical contact region **188** is directly accessible to a suitably configured electrical prong (not shown) that is sized and shaped for insertion into the positive contact slot **172**. A negative battery post **196** and the negative contact slot **174** are enclosed by a negative contact slot encasement **190**. A negative post electrical contact region **198** is directly accessible to a negative electrical prong (not shown) via the negative contact slot **174**.

(75) In an exemplary embodiment, FIG. **10** shows a simplified diagrammatical illustration of a battery **200**, that includes a battery cover **206** and a battery casing **210**, with a positive battery terminal **202** and a negative battery terminal **204** in the battery cover **206**. The battery cover **206** and the battery casing **210** form a battery housing **209**. Included in the battery **200** is a positive contact slot **212** and a negative contact slot **214**, both passing through a battery cover flange seal **208** and through the battery casing **210**. A positive electrical plug **220** may be provided for insertion into the positive contact slot **212** by grasping a plug grip **224** such that the longer dimension of a positive electrical prong **222** is vertically oriented, and such that a positive prong contact surface **226** (hidden) faces the positive battery terminal **202**. Similarly, a negative electrical plug **230** may be grasped by a plug grip **234** such that the longer dimension of a negative electrical prong **232** is vertically oriented for insertion into the negative contact slot **214**, and such that a negative prong contact surface **235** faces the negative battery terminal **204**.

(76) FIG. **11** is a close-up, partial cutaway, detail view of a corner of the battery **200** showing the positive battery terminal **202** with a corner portion of the battery cover **206** removed to expose a positive contact slot encasement **240** and a positive battery post **250**. For clarity of illustration, the thicknesses of the battery cover **206**, the battery cover flange seal **208**, and the battery casing **210** have been exaggerated to enable cross-sectional hatch lines to be seen. In the configuration shown, a positive encasement top surface **242** may be bonded to a battery cover interior surface **228** using a layer of bonding compound (not shown) so as to seal around the positive battery post **250** and prevent battery electrolyte solution (not shown) from seeping into the positive contact slot **212**. In an exemplary embodiment, the positive contact slot encasement **240** may be fabricated as part of the battery cover **206** and thus not require bonding.

(77) The positive contact slot **212** includes a contact slot flange notch **216** that provides an opening in the battery cover flange seal **208**. The positive contact slot **212** also includes a casing contact slot opening **256** in the battery casing **210**. Inside the battery **200**, the positive contact slot **212** includes a positive encasement channel **248** to guide the positive electrical prong **222** (shown in FIG. **10**) to the positive battery post **250**. The positive encasement channel **248** is defined by an encasement channel inner wall **252** and an encasement channel outer wall **254**. The positive electrical prong **222** can be inserted through the contact slot flange notch **216** to make electrical contact with a positive post electrical contact region **258** on a convex surface of the positive battery post **250**. A corner portion of the battery casing **210** has been removed to more clearly show a positive encasement side surface **246** and a positive encasement front surface **244**. Bonding compound or a sealant (not shown) may be applied to either or both of the positive encasement surfaces **244**, **246** during fabrication so as to prevent battery electrolyte solution (not shown) from seeping into the positive contact slot **212** after assembly.

(78) FIG. **12** is a cross-sectional plan view D-D of the battery **200**, taken across the section plane D-D as indicated in FIG. **10**, showing the positive contact slot encasement **240** and a negative contact slot encasement **260**. In this plan view, it can be clearly seen that the positive post electrical contact region **258** is directly accessible to the positive electrical prong **222** (shown in FIG. **10**) when inserted into the positive contact slot **212**, through the battery cover flange seal **208**, through the battery casing **210**, along the positive encasement channel **248**, up to an encasement channel terminus **249**. Note that the encasement channel terminus **249** is slightly curved against the positive battery post **250**. This configuration causes the inserted positive electrical prong **222** to be forced against the positive post electrical contact region **258** when the end of the positive electrical prong **222** is pushed into the encasement groove terminus **249**. This action serves to increase the electrical contact area between the positive electrical prong **222** and the positive battery post **250** to thus reduce contact resistance. Similarly, a negative battery post **262** is partially enclosed in the negative contact slot encasement **260** that includes the negative contact slot **214** with an angled encasement channel terminus **276**. There is a prong stop **274** on an encasement channel wall **272** of the negative contact slot **214**. A negative post electrical contact region **268** is electrically accessible to the negative electrical prong **232**, shown in FIG. **10**, when fully inserted into the negative contact slot **214**.

(79) FIG. **13** is a cross-sectional detail view E-E of the negative battery terminal **204** and the negative contact slot **214** of the battery **200**, taken across the section plane E-E as indicated in FIG. **10**. A negative battery post **262** functions as an electrical conductor between the negative battery terminal **204** and a negative plate strap **282**. An electrolyte-resistant sealant **284** may be applied between the negative contact slot encasement **260** and the battery casing **210** during fabrication to prevent battery electrolyte solution from subsequently seeping into the negative contact slot **214**. The battery cover flange seal **208** may be chemically or thermally bonded to the battery casing **210** during fabrication, as indicated by a battery cover bonding interface **286**. The negative contact slot encasement **260** may also be chemically or thermally bonded to the battery cover **206**, as indicated by a negative encasement bonding interface **288**, or may alternatively be formed as part of the battery cover **206** during fabrication.

(80) Part of the negative battery post **262** protrudes through a post access opening **278** in the encasement channel wall **272** inside the negative contact slot encasement **260** to make the negative post electrical contact region **268** accessible to the negative electrical prong **232** of the negative electrical plug **230**. To provide plug polarity, the negative electrical prong **232** includes a longitudinal prong slot **236** to allow for insertion and withdrawal, indicated by an arrow **238**, of only the negative electrical prong **232** past the prong stop **274** located on the encasement channel wall **272** of the negative encasement channel **264** (seen in FIG. **12**) to the angled encasement channel terminus **276**. The prong stop **274** prevents inadvertent insertion of the positive electrical prong **222**, shown in FIG. **10**, into the negative contact slot **214**. Conversely, the smaller vertical

dimension of the positive contact slot **212** prevents inadvertent insertion of the larger negative electrical prong **232**.

(81) FIG. **14** is a diagrammatical illustration of an alternative design of a positive contact slot encasement **290** as may be used with the battery cover **206** and the positive battery terminal **202**. A corner of the positive contact slot encasement **290** has been truncated to more clearly show a positive encasement cavity **300** bounded by a cavity side wall **296**, a cavity end wall **302**, and a cavity base **304**. The positive battery post **250** is retained in an encasement through-hole **291** that extends from an encasement top surface **298** through the cavity base **304**. It can be seen that prong guidance in the positive contact slot **292** is provided by an encasement groove **294** that is aligned with a corner formed by the cavity side wall **296** and the cavity base **304**. The encasement groove **294** extends past the positive post electrical contact region **258** on the positive battery post **250** into an encasement prong terminus **299**. The positive contact slot encasement **290** is here shown as a separate component configured for bonding to the battery cover **206** with an application of a bonding compound (not shown) to the encasement top surface **298**, or by means of thermal bonding, during battery assembly. The positive contact slot encasement **290** may alternatively be fabricated as an integral part of the battery cover **206**. Bonding compound may also be applied to one or both of an encasement front surface **306** and an encasement side surface **308** for bonding to the battery casing **210** (shown in FIG. **10**).

(82) FIG. **15** is a diagrammatical illustration of another exemplary embodiment of a positive contact slot encasement **310** as may be used with the battery cover **206** (not shown) and the positive battery terminal **202** to enclose a positive contact slot **312** and part of the positive battery post **250**. A corner of the positive contact slot encasement **310** has been truncated to more clearly show an encasement cavity **320** bounded by a cavity side wall **322**, a cavity end wall **324**, and a cavity base **326**. The positive battery post **250** is retained in an encasement through-hole **311** that extends from an encasement top surface **321** through the cavity base **326**. The illustration shows the insertion of the positive electrical prong **222** along the encasement cavity base **326**, past the positive post electrical contact region **258** on the surface of the positive battery post **250**, and into an encasement prong terminus **328**. Insertion and removal of the positive electrical prong **222** is indicated by an arrow **314**. Guidance for the positive electrical prong **222** is by the positive contact slot **312** in a relatively thick positive encasement front wall **318**. Lateral movement (i.e., “wiggling”) of the positive electrical prong **222** during insertion is restricted as the positive contact slot encasement **310** features: (i) the thick positive encasement front wall **318**, providing for a long-length positive contact slot **312**, and (ii) the narrow width of the positive contact slot **312** being sized to provide a sliding friction fit with the inserted positive electrical prong **222**.

(83) FIG. **16** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **340** that enables electrical charging attachment from both the front and the sides of a battery housing **349**. The battery **340** includes a battery cover **346** attached to a battery casing **348** with a battery cover flange seal **368**. There are a positive battery terminal **342** and a negative battery terminal **344** in the battery cover **346**. A positive front contact slot **352** provides an opening in the battery cover flange seal **368** and passes through a casing front face **364**. A positive side contact slot **356** provides a second opening in the battery cover flange seal **368** and passes through a casing left-side face **362**. The battery **340** also includes a negative front contact slot **354** on the casing front face **364**, and a negative side contact slot **358** (hidden) on a casing right-side face **366**.

(84) FIG. **17** is a close-up, partial cutaway, detail view of a corner of the battery **340** showing the positive battery terminal **342**, with a portion of the battery cover **346** removed to expose a positive battery post **350** and a positive contact slot encasement **370**. The positive side contact slot **356** includes an encasement side channel **376** extending to a side slot positive post electrical contact region **378** on the positive battery post **350**. The positive front contact slot **352** includes a positive contact slot flange notch **373** in the battery cover flange seal **368**, with an encasement front channel **372** intersecting at an angle with the encasement side channel **376**, and extending to a front slot

positive post electrical contact region **374** on the positive battery post **350**. A similar configuration of a negative contact slot encasement and contact slot encasement channels (not shown) extending from respective negative contact slots **354**, **358** is provided at the negative battery terminal **344** (shown in FIG. **15**).

(85) FIG. **18** is a detail diagrammatical illustration of the positive battery terminal **102**, the battery cover **106**, the positive contact slot encasement **130**, and the positive battery post **122**, as first shown above in FIGS. **4** and **5**. The positive contact slot **112** is sized and configured to receive a positive electrical plug **380** that includes an electrically conductive positive electrical prong **382**. The positive electrical prong **382** includes a longitudinal alignment ridge **383** that is configured for insertion into a corresponding positive alignment groove **113** in the positive contact slot **112**. This feature prevents erroneous insertion of the positive electrical plug **380** into the negative contact slot **114** (shown in FIG. **4**).

(86) The positive electrical prong **382** also terminates in a leading concave contact edge **386** that is configured to provide maximum contact surface area and minimal electrical contact resistance when inserted against the positive post electrical contact region **128** (shown in FIG. **5**). The positive electrical plug **380** further includes an insulated, high-amperage, electrical conductor **388**, such as stranded copper wire, electrically connected to the positive electrical prong **382** inside an electrically insulative plug grip **384**. The electrical conductor **388** is of sufficient cross-sectional area to safely conduct the electrical current levels normally associated with a battery charging or a jump-starting operation.

(87) A negative electrical plug **390** includes an electrically conductive negative electrical prong **392** that is a mirror image of the positive electrical prong **382**, and is thus configured for insertion into the negative contact slot **114** (shown in FIG. **4**). An alignment ridge **393** on the negative electrical prong **392** prevents erroneous insertion of the negative electrical plug **390** into the positive contact slot **112**. The negative electrical prong **392** terminates in a leading concave contact edge **396** which is configured to fit against the negative post electrical contact region **148** (shown in FIG. **6**). The negative electrical plug **390** is connected to an insulated, high-amperage, electrical conductor **398** inside an electrically insulative plug grip **394**.

(88) FIG. **19** is a simplified diagrammatical illustration of the positive battery terminal **102** and the negative battery terminal **104** on the battery cover **106**, first seen in FIG. **4**. In an alternative embodiment, the positive contact slot **112** is replaced by a positive contact slot **402** which has a substantially rectangular opening with a truncated upper left-hand corner **403**. The negative contact slot **114** is replaced by a mirror-image negative contact slot **404** which has a substantially rectangular opening with a truncated upper right-hand corner **405**.

(89) A positive electrical plug **410** includes a positive electrical prong **412** configured for insertion only into the positive contact slot **402**. An insulated electrical conductor **418** is connected to the positive electrical prong **412** inside a plug grip **414**. A polarizing flat **413** on the positive electrical prong **412** provides clearance for the truncated upper left-hand corner **403**, and allows a user to grasp the plug grip **414** and place a prong concave contact surface **416** against the positive battery post **122** enclosed within a positive contact slot encasement **406**. A negative electrical plug (not shown), having a negative electrical prong that is a mirror image of the positive electrical prong **412**, can be used for insertion only into the polarized negative contact slot **404** in a negative contact slot encasement **408**.

(90) FIG. **20** is a simplified diagrammatical illustration of a portion of a battery **420** wherein electrical connection to the positive battery terminal **102** can be made through a substantially rectangular positive contact slot **422** that extends to the positive battery post (not shown). There is also a substantially rectangular negative contact slot **424** providing electrical contact access to the negative battery terminal **104** via a negative battery post (not shown). The positive contact slot **422** includes a perimeter ridge **426** functioning to prevent insertion of an incompatible electrical prong (not shown) that does not have a corresponding channel of a proper size and location. The negative

contact slot **424** similarly includes a perimeter ridge **428** to prevent insertion of an incompatible electrical prong (not shown). The contact slots **422**, **424** thus form a polarized contact slot pair.

(91) FIG. **21** is a simplified diagrammatical illustration of a portion of a battery **430** wherein an electrical connection to the positive battery terminal **102** can be made through a substantially rectangular positive contact slot **432** that extends to the positive battery post (not shown). Similarly, electrical contact access to the negative battery terminal **104** can be made via a substantially rectangular negative contact slot **434** that extends to the negative battery post (not shown). As can be seen from the illustration, the dimensional width of the positive contact slot **432** is greater than that of the negative contact slot **434**, and the height dimension of the negative contact slot **434** is greater than that of the positive contact slot **432**, thus forming a polarized contact slot pair.

(92) FIG. **22** is a simplified diagrammatical illustration of a portion of a battery **440** wherein an electrical connection can be made to the positive battery terminal **102** via an offset jog rectangular positive contact slot **442** extending through the battery cover **106** and the battery casing **110**. Similarly, electrical contact access to the negative battery terminal **104** is provided by an offset jog rectangular negative contact slot **444**. The positive contact slot **442** is a mirror image of the negative contact slot **444**, to form a polarized contact slot pair.

(93) FIG. **23** is a simplified diagrammatical illustration of a contact slot encasement **450** that can be used in place of either of the contact slot encasements **130**, **140**, shown in FIG. **6**, and in place of either of the contact slot encasements **180**, **190**, shown in FIG. **9**. The contact slot encasement **450** has a rectangular perimeter with a first encasement wall **452**, a second encasement wall **454**, and a prong support base **456**. A battery post through hole **466** passes through the encasement walls **452**, **454** and the prong support base **456**. A base through hole opening **468** is sized and shaped to allow insertion of a battery post (not shown), while providing a leakproof fit around a battery post (not shown) to prevent the seepage of battery electrolyte solution onto the prong support base **456**. The contact slot encasement **450** may be attached to the battery cover interior surface (not shown) by bonding a top surface **458** to the battery cover, or by applying an adhesive to the top surface **458**. An electrolyte-resistant sealant (not shown) may be applied to a front surface **462** and to a side surface **464** to prevent seepage of battery electrolyte (not shown) onto the prong support base **456**.

(94) FIG. **24** is a simplified diagrammatical illustration of the battery terminals **102**, **104**, **162**, **164** with respective battery posts **122**, **142**, **186**, **196**. When used with any of contact slot encasements **130**, **140**, **180**, or **190** described above, a respective post electrical contact region **128**, **148**, **188**, **198** is exposed as an electrical contact for a compatible electrical plug, such as the positive electrical plug **380** in FIG. **18** and the positive electrical plug **410** in FIG. **19**. In FIG. **24**, the positive post electrical contact region **128** is a nonplanar rectangle, or a curved band, bounded by a first circumferential arc **472**, a second circumferential arc **474**, a first longitudinal line segment **476**, and a second longitudinal line segment **478**. The positive post electrical contact region **128** interfaces with external electrical connectors such as the positive electrical plug **380** (shown in FIG. **18**) and the positive electrical plug **410** (shown in FIG. **19**). It should be understood that, although the positive electrical plug **380** has the prong concave contact surface **386**, and the positive electrical plug **410** has the prong concave contact surface **416**, contact surfaces of larger radii (not shown) can be used instead, under certain circumstances, but a greater contact resistance can result.

(95) FIG. **25** is a simplified diagrammatical illustration of a contact slot encasement **480** configuration that can be used in place of either of the contact slot encasements **240**, **260** shown in FIG. **12**, the positive contact slot encasement **290** shown in FIG. **14**, the positive contact slot encasement **310**, shown in FIG. **15**, and the positive contact slot encasement **370**, shown in FIG. **17**. The contact slot encasement **480** has a rectangular perimeter with a first encasement wall **482**, a second encasement wall **484**, and an encasement interior pad **486**. A battery post through hole **490** passes through the encasement walls **482**, **484** and through an encasement base surface **488**. A base through hole opening **498** is sized and shaped to allow insertion of a battery post, while providing a

leakproof fit around a battery post (not shown) to prevent the seepage of battery electrolyte solution into a first prong groove **492** or a second prong groove **494**. The contact slot encasement **480** may be attached to a battery cover (not shown) by bonding an encasement top surface **496** to the battery cover interior surface, or by applying an adhesive (not shown) to the encasement top surface **496** for chemical bonding. An electrolyte-resistant sealant (not shown) may be applied to an encasement first side surface **483** and to an encasement second side surface **485** to prevent seepage of battery electrolyte (not shown) into the prong grooves **492**, **494**.

(96) FIG. **26** shows the battery terminals **202**, **204**, **342** and the corresponding battery posts **250**, **262**, **350**, as disclosed above. The electrical contact regions **258**, **268**, **374**, **378** are nonplanar rectangles, or curved bands, bounded by a first circumferential arc **502**, a second circumferential arc **504**, a first longitudinal line segment **506**, and a second longitudinal line segment **508**. The electrical contact regions **258**, **268**, **374**, **378** interface with external electrical connectors such as the electrical plugs **220**, **230** in FIG. **10**.

(97) FIG. **27** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **510** that enables electrical charging access in a battery cover **516**. The battery **510** includes a positive battery terminal **512** and a negative battery terminal **514**. A positive contact slot **522** in the battery cover **516**, is located between the positive battery terminal **512** and a cover perimeter front edge **518**, and provides a means of electrically connecting to the positive battery terminal **512**. A negative contact slot **524** in the battery cover **516**, is located between the negative battery terminal **514** and the cover perimeter front edge **518**, and provides a means of electrically connecting to the negative battery terminal **514**.

(98) FIG. **28** is a cross-sectional detail view F-F of the positive battery terminal **512** and the positive contact slot **522** of the battery **510**, taken across the section plane F-F through the battery cover **516**, as indicated in FIG. **27** except that, for clarity of illustration, the section plane F-F does not pass through the battery terminal **512**, nor through a positive battery post **526**. Note that a longitudinal axis **523** of the positive contact slot **522** forms an acute angle 'G' with a longitudinal axis **527** of the positive battery post **526**. A positive contact slot encasement **520** encloses the positive contact slot **522** and part of the positive battery post **526**.

(99) As can be appreciated by one skilled in the relevant art, the battery **510** can be charged, or can be used to electrically connect to another battery, by insertion of a positive electrical plug **530** having a slot-compatible positive electrical prong **532** into the positive contact slot **522**, as indicated by an arrow **571**. This action makes electrical contact with a convex positive post electrical contact region **528**, which region is in the shape of a curved band on the positive battery post **526**. The positive electrical prong **532** is electrically connected to an insulated electrical conductor **538** inside a non-conducting plug grip **536**. It should be understood that all the electrical prongs disclosed in the present specification are substantially in the shape of a rectangular parallelepiped with one rectangular edge secured in a plug grip. As best seen in the detail view of the positive electrical plug **530** in FIG. **29**, the electrical prong **532** includes a top surface **543**, a bottom surface **545** (hidden), a right edge **547**, and a left edge **549**. The positive electrical prong **532** terminates in a leading oblique concave contact edge **533** for achieving maximum surface contact with the convex positive post electrical contact region **528**, shown in FIG. **28**. That is, the longitudinal length of the top surface **543** is smaller than the longitudinal length of the bottom surface **545**.

(100) FIG. **30** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **550**, with a positive battery terminal **552** and a negative battery terminal **554**, that enables electrical charging attachment on a battery cover front perimeter edge **558**. The battery **550** includes a positive contact slot **562** in a positive recessed surface **566** that provides a means of electrically connecting to the positive battery terminal **552**. A negative contact slot **564** in a negative recessed surface **568** provides a means of electrically connecting to the negative battery terminal **554**.

(101) FIG. **31** is a simplified diagrammatical illustration of an exemplary embodiment of a battery

570 that provides electrical charging attachments in a battery cover **571**. A positive contact slot **582**, in a positive recessed surface **586**, is located in a cover left perimeter edge **576** to provide a means of electrically connecting to a positive battery terminal **572**. A negative contact slot **584** (hidden), in a negative recessed surface **588**, is located in a cover right perimeter edge **578** and provides a means of electrically connecting to a negative battery terminal **574**.

(102) FIG. **32** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **590** that provides electrical charging attachments in a battery casing **591**. A positive contact slot **602** (hidden), in a positive recessed surface **606**, is located in a left casing corner **596** and provides a means of electrically connecting to a positive battery terminal **592**. A negative contact slot **604** in a negative recessed surface **608**, is located in a right casing corner **598** and provides a means of electrically connecting to a negative battery terminal **594**.

(103) FIG. **33** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **610** that provides electrical charging attachments on corners of a battery cover **620**. In this configuration, a positive corner facet **622** is provided at the intersection of a cover perimeter left edge **624** and a cover perimeter front edge **628**. A negative corner facet **632** is provided at the intersection of a cover perimeter right edge **634** and the cover perimeter front edge **628**. A positive contact slot **616** in the positive corner facet **622** provides a means of electrically connecting to a positive battery terminal **612**. A negative contact slot **618** (hidden) in the negative corner facet **632** provides a means of electrically connecting to a negative battery terminal **614**.

(104) FIG. **34** is an alternate cross-section A-A plan view of an exemplary embodiment of the battery **100** of FIG. **6** showing an alternative contact slot encasement configuration. In this configuration, the positive contact slot **112** and the positive post electrical contact region **128** are enclosed in a positive contact slot encasement **636**, wherein the positive contact slot encasement **636** covers about half the circumference of the positive battery post **122**. Similarly, the negative contact slot **114** and the negative post electrical contact region **148** are enclosed in a negative contact slot encasement **638**, wherein the negative contact slot encasement **638** covers about half the circumference of the negative battery post **142**.

(105) It can be appreciated by one skilled in the relevant art that the above-disclosed configurations of contact slot encasements covering respective battery posts over their entire circumferences provides can for minimal leakage of battery electrolyte solution. However, innovative battery manufacturing methods of sealing component interfaces may provide for minimal leakage of electrolyte solution even if less than the entire circumference of a battery post is covered. FIG. **35** is an alternate cross-section C-C plan view of an exemplary embodiment of the battery **160** of FIG. **9** showing an alternative contact slot encasement configuration. In this configuration, the positive contact slot **172** and the positive post electrical contact region **188** are enclosed in a positive contact slot encasement **650**, wherein the positive contact slot encasement **650** covers more than half the circumference of the positive battery post **186**. Similarly, the negative contact slot **174** and the negative post electrical contact region **198** are enclosed in a negative contact slot encasement **651**, wherein the negative contact slot encasement **651** covers more than half the circumference of the negative battery post **198**.

(106) FIG. **36** is an alternate cross-section D-D plan view of an exemplary embodiment of the battery **200** of FIG. **12** showing an alternative contact slot encasement configuration. In this configuration, the positive contact slot **212** and the positive post electrical contact region **258** are enclosed in a positive contact slot encasement **660**, wherein the positive contact slot encasement **660** covers less than half the circumference of the positive battery post **250**. Similarly, the negative contact slot **214** and the negative post electrical contact region **268** are enclosed in a negative contact slot encasement **661**, wherein the negative contact slot encasement **661** covers less than half the circumference of the negative battery post **262**.

(107) But wait, there is more. FIG. **37** is a simplified diagrammatical illustration of a battery **640** having a positive battery side terminal **642** and a negative battery side terminal **644** located on a

battery casing front surface **646** of a battery casing **648**. A positive top electrical plug **662** can be inserted into a positive top contact slot **652** to make electrical contact with the positive battery side terminal **642**. The positive top electrical plug **662** includes a positive top electrical prong **664**, with a positive top prong leading edge concave contact surface **666**, connected to an insulated electrical conductor **668**. Similarly, a negative top electrical plug **656** can be inserted into a negative top contact slot **654** to make electrical contact with the negative battery side terminal **644**. The top contact slots **652**, **654** are located in a battery cover **658**.

(108) In an exemplary embodiment, a positive side contact slot **672** may be provided in a battery casing left-side surface **676**, and a negative side contact slot **674** (hidden) may be provided in a battery casing right-side surface **678** (hidden). A positive side electrical plug **682** can be inserted into the positive side contact slot **672**, and a negative side electrical plug **684** can be inserted into the negative side contact slot **674**. The positive side electrical plug **682** includes a positive side electrical prong **686**, with a positive side prong leading edge concave contact surface **688**, connected to an insulated electrical conductor **689**. The negative side electrical plug **684** is similarly configured. It should be understood that the battery **640** need include only one of the positive contact slots **652**, **672** and only one of the negative contact slots **654**, **674** when the battery **640** is manufactured. Alternatively, each of the contact slots **652**, **654**, **672**, **674** may be covered by a knock-out tab (not shown) that can be removed by a user after the battery **640** has been installed in an engine compartment.

(109) FIG. **38** is a close-up, partial cutaway, detail view of the positive battery side terminal **642** connected to a positive battery post **692** on the battery casing **648**. Part of a positive contact slot encasement **690** has been removed to expose the partially enclosed positive battery post **692**. The particular method by which the positive battery post **692** is attached to a positive plate strap (not shown) lying below the positive contact slot encasement **690**, can vary from manufacturer to manufacturer and is not shown, for clarity of illustration. The positive side electrical plug **682** is shown as partially inserted through the positive side contact slot **672** in the battery casing left-side surface **676**.

(110) The positive top electrical prong **664** is electrically connected to the insulated electrical conductor **668** inside a top positive plug grip **667**. The positive top prong leading edge concave contact surface **666** has the same radius of curvature as does the outer surface of the positive battery post **692**. Accordingly, the positive top electrical plug **662** can be used to make electrical contact with a positive top slot electrical contact region **694** on the positive battery post **692** by full insertion into the positive top contact slot **652** (shown in FIG. **34**). Similarly, the positive side prong leading edge concave contact surface **688** (also shown in FIG. **34**) on the positive side electric prong **686** has the same radius of curvature as does the outer surface of the positive battery post **692**. Accordingly, the positive side electrical plug **682** can be used to make electrical contact with a positive side slot electrical contact region **696**.

(111) FIG. **39** shows the positive battery post **692** and the positive battery side terminal **642** in greater detail. The positive top slot electrical contact region **694** is a nonplanar rectangle, or a curved band, on the convex cylindrical surface of the positive battery post **692**, bounded by a first circumferential arc **702**, a second circumferential arc **704**, a first longitudinal line segment **706**, and a second longitudinal line segment **708**. The positive top slot electrical contact region **694** interfaces with external electrical connectors such as the top electrical plugs **662**, **656** shown in FIG. **34**.

(112) FIG. **40** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **710** that enables electrical charging attachment at both the top and the sides of a battery casing **711**, in accordance with the present invention. A positive top contact slot **716** provides an opening in a battery cover **722**, and a positive side contact slot **717** provides an opening in a battery casing left-side surface **726**. The positive contact slots **716**, **717** provide alternate modes of electrically contacting a positive battery side terminal **712**. The battery **710** also includes a negative top contact

slot **718** in the battery cover **722**, and a negative side contact slot **719** (hidden) in a battery casing right-side surface **728** (hidden). The negative contact slots **718**, **719** provide alternate modes of electrically contacting a negative battery side terminal **714**.

(113) A positive top electrical plug **736** can be inserted into the positive top contact slot **716**, or a positive side electrical plug **737** can be inserted into the positive side contact slot **717**, to make electrical contact with the positive battery side terminal **712**. A negative top electrical plug **738** can be inserted into the negative top contact slot **718**, or a negative side electrical plug **739** can be inserted into a negative side contact slot **719** (hidden), to make electrical contact with the negative battery side terminal **714**. It should be understood that the battery **710** needs to include only positive contact slot **716** or **717**, and only negative contact slot **718** or **719**, to enable a charging operation. Accordingly, each of the contact slots **716**, **717**, **718**, **719** may be covered by a knock-out tab (not shown) that can be removed by a user after the battery **710** has been installed in an engine compartment.

(114) FIG. **41** is a close-up, partial cutaway, detail view of a portion of the battery casing left-side surface **726** and the positive battery side terminal **712**, which is connected to a positive battery post **740** partially secured in a positive contact slot encasement **730**. Part of the positive contact slot encasement **730** has been removed to more clearly show part of the positive battery post **740**. The particular method of attaching the positive battery post **740** to a positive plate strap (not shown), lying below the positive contact slot encasement **730**, varies from manufacturer to manufacturer and is not shown, for clarity of illustration.

(115) The positive top electrical plug **736** includes a positive top electrical prong **742** with a positive top plug contact surface **744** (hidden) that makes electrical contact with a positive top slot electrical contact region **746** on the positive battery post **740** when fully inserted into the positive top contact slot **716** (shown in FIG. **37**). The positive side electrical plug **737** includes a positive side electrical prong **743** with a positive side plug contact surface **745** (hidden) that makes electrical contact with a positive side slot electrical contact region **747** on the positive battery post **740** when fully inserted into the positive side contact slot **717** in the battery casing left-side surface **726**. It should be noted that the electrical prongs **742**, **743** need not be as thick as the electrical prongs **664**, **686** shown in FIG. **35**. It can be appreciated by one skilled in the art that, because the electrical prongs **664**, **686** make electrical contact at respective prong leading edge concave contact surfaces **666**, **688**, the thicker configuration results in smaller contact resistance at the respective electrical contact regions **746**, **747**.

(116) FIG. **42** shows the positive battery post **740** and the positive battery side terminal **712**. The positive top slot electrical contact region **746** is a nonplanar rectangle, or a curved band, on a convex surface bounded by a first circumferential arc **752**, a second circumferential arc **754**, a first longitudinal line segment **756**, and a second longitudinal line segment **758**. The positive top slot electrical contact region **746** interfaces with external electrical connectors such as the positive top electrical plug **736**, in FIG. **38**.

(117) FIG. **43** is a simplified diagrammatical illustration of an exemplary embodiment of a battery **760** that provides electrical charging attachments in a battery cover **761**. The battery **760** has a positive battery side terminal **762** and a negative battery side terminal **764** in a battery casing front surface **765**. A positive electrical contact slot **766** extends into a positive slot encasement **772** and a negative electrical contact slot **768** extends into a negative slot encasement **774**, both slot encasements **772**, **774** attached to a battery casing **763** to prevent leakage of battery electrolyte solution (not shown) into the contact slots **766**, **768**. A positive electrical plug **770** has a positive electrical prong **776** that is configured for insertion only into the positive electrical contact slot **766** and the positive slot encasement **772**, by which action an electrical connection can be made to the positive battery side terminal **762**. The negative electrical contact slot **768** and the negative slot encasement **774** are similarly configured for insertion of only a negative electrical plug (not shown).

(118) As seen in FIG. 44, the positive battery side terminal **762** is connected to a positive battery post **782** which is, in turn, electrically connected to an internal positive battery plate (not shown). The illustration shows relative positions of the positive electrical plug **770** and the positive battery side terminal **762** as the positive electrical plug **770** is being inserted into the positive electrical contact slot **766** (shown in FIG. 40). The positive electrical prong **776** includes a prong top surface **773**, a prong contact surface **775** (hidden), a prong right edge **777**, and a prong left edge **779**. The positive electrical prong **776** terminates in a prong leading edge **778**. When the positive electrical plug **770** is fully inserted into the positive electrical contact slot **766**, the prong contact surface **775** makes electrical contact with a positive post electrical contact region **788** on the positive battery post **782**.

(119) FIG. 45 illustrates an exemplary embodiment of the positive slot encasement **772** which includes an encasement side wall **792**, an encasement back wall **794**, and an encasement base **796** defining an encasement cavity **798**. The encasement back wall **794** includes a cavity convex surface **806** inside the positive slot encasement **772**. The encasement base **796** includes a base notch **808** to provide clearance for the positive battery post **782** (shown in FIG. 44). A front encasement surface **802** and a side encasement surface **804** are attached to the inside of the battery casing **763** (shown in FIG. 43). As the positive electrical plug **770** is inserted into the encasement cavity **798**, the prong top surface **773** (shown in FIG. 44) engages the cavity convex surface **806**, whereby the prong bottom surface **775** is urged against the positive post electrical contact region **788** (both shown in FIG. 44). The negative slot encasement **774** is similarly configured, as substantially a mirror image of the positive slot encasement **772**.

(120) FIG. 46 is an exemplary embodiment of an alternative positive slot encasement **810** that may be used in the battery **760** in place of the positive slot encasement **772**. The positive slot encasement **810** includes an encasement side wall **812**, an encasement back wall **814**, and an encasement base **816** defining an encasement cavity **818**. An encasement front surface **822** and an encasement side surface **824** are attached to the inside of the battery casing **763** (shown in FIG. 43) to prevent seepage of internal battery electrolyte solution (not shown) into the encasement cavity **818**. The encasement back wall **814** includes a cavity sloping surface **826** for engaging the prong leading edge **778** on the positive electrical prong **776** (both shown in FIG. 44), and the encasement base **816** includes a base through slot **828** that allows the positive battery post **782** (not shown) to pass through the encasement base **816**.

(121) FIG. 47 is a simplified diagrammatical illustration of an exemplary embodiment of a battery **830** that enables electrical charging attachment in a battery casing **831**. A positive electrical contact slot **836** extends into a positive slot encasement **842** and a negative electrical contact slot **838** (hidden) extends into a negative slot encasement **844**. A positive electrical plug **840** is configured for insertion into only the positive electrical contact slot **836** and the positive slot encasement **842**, by which action an electrical connection can be made to a positive battery side terminal **832**. The negative electrical contact slot **838** and the negative slot encasement **844** (a mirror image of positive slot encasement **842**) are similarly configured for insertion of a negative electrical plug (not shown) to electrically engage a negative battery side terminal **834**.

(122) As seen in FIG. 48 the positive battery side terminal **832** is connected to an angular positive battery post **852** which has a post transition section **854** extending into the battery **830** (shown in FIG. 47) and a post depending leg **856** that extends to and is electrically connected to a positive battery plate (not shown) disposed below the positive slot encasement **842** (shown in FIG. 47). The illustration shows relative positions of the positive electrical plug **840** and the positive battery side terminal **832** as the positive electrical plug **840** is being inserted into the positive electrical contact slot **836** (shown in FIG. 47). When the positive electrical plug **840** is fully inserted into the positive electrical contact slot **836**, a prong contact surface **848** (hidden) on a positive electrical prong **846** makes electrical contact with a positive post electrical contact region **858** on the post transition section **854**.

(123) FIG. 49 illustrates an exemplary embodiment of the positive slot encasement **842** which is configured for use with the positive battery post **852** (shown in FIG. 48). The positive slot encasement **842** includes an encasement side wall **862**, an encasement back wall **864**, an encasement base **866**, and an encasement top **868** defining an encasement cavity **872**. The encasement top **868** includes a cavity convex surface **874** inside the positive slot encasement **842**, and a through slot **876** to allow the post depending leg **856** (shown in FIG. 48) to contact the positive battery plate (not shown). With additional reference to FIG. 48, as the positive electrical plug **840** is inserted into the encasement cavity **872**, a leading edge **849** on the positive electrical prong **846** engages the cavity convex surface **874**, whereby the prong contact surface **848** is urged against the positive post electrical contact region **858**. The negative slot encasement **844** is similarly configured, but is substantially a mirror image of the positive slot encasement **842**.

(124) FIG. 50 illustrates a typical application of the present invention in the process of charging of a battery, or in the jump-starting of a vehicle, in which a source battery **880** and a drained battery **890** have the same contact slot configurations. This charging situation may occur, for example, when both vehicles are made by the same manufacturer. The source battery **880** includes a positive contact slot **886** for electrically connecting to a positive battery terminal **882**, and a negative contact slot **888** for electrically connecting to a negative battery terminal **884**. The drained battery **890** includes a positive contact slot **896** (hidden) for electrically connecting to a positive battery terminal **892**, and a negative contact slot **898** (hidden) for electrically connecting to a negative battery terminal **894**. It should be understood that, to prevent inadvertent errors in connection, the opening size and shape of the positive contact slot **886** is different from the opening size and shape of the negative contact slot **888**, and the opening size and shape of the positive contact slot **896** is different from the opening size and shape of the negative contact slot **898**.

(125) A battery interconnection cable assembly **900** is used to electrically connect the source battery **880** to the drained battery **890**, as shown. The battery interconnection cable assembly **900** includes a source battery connection adapter **902** used at the source battery **880**, a drained battery connection adapter **903** used at the drained battery **890**, and an adapter connection cable **910** used to interconnect the battery connection adapters **902**, **903**. The source battery connection adapter **902** is electrically connected to the source battery **880** by inserting a positive source electrical plug **906** into the positive contact slot **886** and a negative source electrical plug **908** into the negative contact slot **888**. The drained battery connection adapter **903** is electrically connected to the drained battery **890** by inserting a positive drained electrical plug **907** into the positive contact slot **896** and a negative drained electrical plug **909** into the negative contact slot **898**.

(126) The adapter connection cable **910** includes a source polarized plug **912** and a drained polarized plug **913** at opposite ends of a cable conductor pair **914** that includes a positive insulated conductor **916** and a negative insulated conductor **918**. Preferably, the cable conductor pair **914** is approximately three to six meters in length to enable electrically connecting batteries in adjacent vehicles. The electrical connection between the source battery **880** and the drained battery **890** may be completed by inserting the source polarized plug **912** into a source polarized receptacle **904** on the source battery connection adapter **902** and by inserting the drained polarized plug **913** into a drained polarized receptacle **905** on the drained battery connection adapter **903**. In an exemplary embodiment, the polarized plugs **912**, **913** may be interchangeable so that the adapter connection cable **910** is reversible end-for-end. In the charging operation shown in FIG. 50, the driver of the vehicle with the source battery **880** may typically provide the source battery connection adapter **902** and the adapter connection cable **910**. The driver of the vehicle with the drained battery **890** may typically provide the drained battery connection adapter **903** and may also have his own adapter connection cable **910**.

(127) It should be understood that the source battery **880** may include contact slots of a size and shape other than those shown for the positive contact slot **886** and the negative contact slot **888**. For example, any of the positive contact slot **112** shown in FIG. 4, the positive contact slot **172**

shown in FIG. 8, the positive contact slot 212 shown in FIG. 10, the positive contact slot 402 shown in FIG. 19, or the positive contact slot 422 shown in FIG. 20, can be used in place of the positive contact slot 886 in FIG. 50. Accordingly, the negative contact slot 888 and the electrical plugs 906, 908 would be replaced by a polarized negative contact slot and electrical plugs of compatible configurations.

(128) FIG. 51 is a detailed view of the source battery connection adapter 902. The source polarized receptacle 904 is a nonconductive enclosure that includes a positive electrical socket 922 connected to the positive source electrical plug 906 by a positive insulated conductor 926, and a negative electrical socket 924 connected to the negative source electrical plug 908 by a negative insulated conductor 928. It can be appreciated that the positive source electrical plug 906 and the negative source electrical plug 908 are noninterchangeable such that the positive source electrical plug 906 cannot be inserted into the negative contact slot 888, and the negative source electrical plug 908 cannot be inserted into the positive contact slot 886, shown in FIG. 50.

(129) FIG. 52 is a simplified detail illustration of the adapter connection cable 910 showing that the source polarized plug 912 includes a positive electrical prong 932 electrically connected to a positive electrical prong 933 in the drained polarized plug 913 by the positive insulated conductor 916, and that a negative electrical prong 934 is electrically connected to a negative electrical prong 935 in the drained polarized plug 913 by the negative insulated conductor 918. The polarized plugs 912, 913 include respective flexible, nonconductive hoods 914, 915 that can be folded back to fold lines 917, 919 if desired to allow insertion of the electrical prongs 932, 933, 934, 935 into the polarized receptacles 904, 905 (shown in FIG. 50), or into alternatively configured receptacles (not shown). The positive insulated conductor 916 includes a positive electrical conductor 936, and the negative insulated conductor 918 includes a negative electrical conductor 938. Each of the electrical conductors 936, 938 has a sufficient cross-sectional area, (e.g., preferably four gauge to zero gauge), to safely conduct high amperage engine starter current.

(130) FIG. 53 shows a battery adapter cable 940 that can be used in place of both the source battery connection adapter 902 and the adapter connection cable 910 to perform the battery charging operation shown in FIG. 50. Referring to FIG. 53, the battery adapter cable 940 includes the positive source electrical plug 906, which is electrically connected to the positive insulated conductor 926, to the positive insulated conductor 916, and to the positive electrical prong 933 in the drained polarized plug 913. The battery adapter cable 940 also includes the negative source electrical plug 908, which is electrically connected to the negative insulated conductor 928 to the negative insulated conductor 918, and to the negative electrical prong 935 in the drained polarized plug 913. Inside a nonconductive electrical junction casing 944, the positive insulated conductor 926 is electrically connected to the positive insulated conductor 916, and the negative insulated conductor 928 is electrically connected to the negative insulated conductor 918.

(131) FIG. 54 shows a battery interconnection cable 950 that can be used in place of the battery interconnection cable assembly 900 shown in FIG. 50. Referring to FIG. 54, the battery interconnection cable 950 includes the positive source electrical plug 906, which is electrically connected to the positive insulated conductor 926, to the positive insulated conductor 916, to the positive insulated conductor 927, and to the positive drained electrical plug 907. The battery interconnection cable 950 also includes the negative source electrical plug 908, which is electrically connected to the negative insulated conductor 928, to the negative insulated conductor 918 to the negative insulated conductor 929, and to the negative drained electrical plug 909. The positive insulated conductor 926 is electrically connected to the positive insulated conductor 916, and the negative insulated conductor 928 is electrically connected to the negative insulated conductor 918, inside an electrical junction casing 954. The positive insulated conductor 916 is electrically connected to the positive insulated conductor 927 and the negative insulated conductor 918 is electrically connected to the negative insulated conductor 929, inside an electrical junction casing 955.

(132) The particular configurations of the polarized plugs and the polarized sockets used in mating an adapter connection cable with battery connection adapters can be similar to any one of many polarized configurations widely used in the relevant art. For example, a battery connection adapter **960** may use a polarized receptacle **966** having a positive electrical socket **962** with a groove **963** and a negative electrical socket **964** without a groove, as shown in FIG. 55. A battery connection adapter **970** may use a polarized receptacle **976** having a positive electrical socket **972** oriented at an angle, i.e., not parallel, relative to a negative electrical socket **974**, as shown in FIG. 56. A battery connection adapter **980** may use a polarized receptacle **986** having a round positive electrical socket **982** and a smaller round negative electrical socket **984**, as shown in FIG. 57. A battery connection adapter **990** may use a coaxial polarized receptacle **996** having a central conductive positive sleeve **992** and an outer conductive negative cylindrical shell **994**, as shown in FIG. 58.

(133) A can be appreciated by one skilled in the art, a polarized plug (not shown) used in an adapter connection cable for attachment to any of the polarized receptacles **966**, **976**, and **986** would include appropriately configured positive and negative electrical prongs in a plug housing such that the positive electrical prong fits into the positive electrical socket and the negative electrical prong fits into the negative electrical socket. A polarized plug (not shown) used in an adapter connection cable for attachment to the coaxial polarized receptacle **996** would have a central conductive pin configured for contact insertion into the central conductive positive sleeve **992**, and a larger cylindrical conductor configured for frictional fit inside the negative cylindrical shell **994**.

(134) FIG. 59 illustrates a typical application of the present invention in the process of charging of a battery, or in the jump-starting of a vehicle, in which the source battery **880** and a drained battery **1000** have different contact slot configurations. This charging situation may occur, for example, when the vehicles are made by different manufacturers. In the example shown, the drained battery **1000** includes a positive battery side terminal **1002** and a negative battery side terminal **1004**. Electrical connections to the battery side terminals **1002**, **1004** are provided by a positive contact slot **1006** and a negative contact slot **1008**, respectively. A drained battery connection adapter **1010** is used with the adapter connection cable **910** and the source battery connection adapter **902** to electrically connect the source battery **880** to the drained battery **1000**.

(135) The drained battery connection adapter **1010** includes the drained polarized receptacle **905** with a positive electrical socket **923** connected to a positive drained electrical plug **1012** by a positive insulated conductor **1016**, and a negative electrical socket **925** connected to a negative drained electrical plug **1014** by a negative insulated conductor **1018**. The positive drained electrical plug **1012** and the negative drained electrical plug **1014** are noninterchangeable such that the positive drained electrical plug **1012** cannot be inserted into the negative contact slot **1008**, and the negative drained electrical plug **1014** cannot be inserted into the positive contact slot **1006**. In an exemplary embodiment, the battery adapter cable **940**, shown in FIG. 53, can be used in place of the source battery connection adapter **902** and the adapter connection cable **910**. In the charging operation shown in FIG. 59, the driver of the vehicle with the source battery **880** may typically provide the source battery connection adapter **902** and the adapter connection cable **910**. The driver of the vehicle with the drained battery **1000** may typically provide the drained battery connection adapter **1010** and may also have his own adapter connection cable **910**.

(136) FIG. 60 illustrates another typical application of the present invention in the process of charging of a battery, or in the jump-starting of a vehicle, in which a drained battery **1020** has a positive battery side terminal **1022** and a negative battery side terminal **1024**, but no contact slots. In the example shown, a user has installed on the drained battery **1020** a battery side mount clamp **1027** that includes a positive contact slot **1026** electrically connected to the positive battery side terminal **1022**, and a negative contact slot **1028** electrically connected to the negative battery side terminal **1024**. Various embodiments of the battery side mount clamp **1027** are disclosed in U.S. Pat. No. 10,862,231 incorporated by reference herein in its entirety. The source battery **880** is

electrically connected to the drained battery **1020** with a battery side mount clamp connection adapter **1030** and the battery adapter cable **940**.

(137) The battery side mount clamp connection adapter **1030** includes a positive electrical socket **1032** electrically connected to a positive electrical prong **1036**, and a negative electrical socket **1034** electrically connected to a negative electrical prong **1038**. Electrical connections are disposed inside a nonconductive connection adapter grip **1035**. The electrical sockets **1032**, **1034** are sized and shaped so as to enable insertion therein of the respective electrical prongs **933**, **935** in the battery adapter cable **940**, shown in FIG. 50. In the charging operation shown in FIG. 60, the driver of the vehicle with the source battery **880** may typically provide the battery adapter cable **940**. The driver of the vehicle with the drained battery **1020** may typically provide the battery side mount clamp connection adapter **1030**.

(138) FIG. 61 illustrates an exemplary application of the present invention in the process of charging of a battery, or in the jump-starting of a vehicle, in which a drained battery **1060** has a positive battery terminal **1062**, a negative battery terminal **1064**, but no contact slots. In this charging configuration, a polarized battery clamp **1070** may be used to electrically access the drained battery **1060**. The polarized battery clamp **1070** is disclosed in U.S. Pat. No. 9,815,378, incorporated by reference herein in its entirety. In an alternative embodiment, a compact battery clamp (not shown) disclosed in U.S. Pat. No. 9,178,292, also incorporated by reference herein in its entirety, can be used in place of the polarized battery clamp **1070**.

(139) A source battery **1040** with a positive battery post **1042** and a negative battery post **1044** may be electrically connected to the polarized battery clamp **1070** on the drained battery **1060** using a source battery connection adapter cable **1050**. The source battery connection adapter cable **1050** includes: (i) the drained polarized plug **913**, (ii) the insulated conductors **916**, **918**, (iii) a positive source electrical plug **1052** with a positive electrical prong **1056** electrically connected to a positive insulated conductor **1053**, and (iv) a negative electrical plug **1054** with a negative electrical prong **1058** electrically connected to a negative insulated conductor **1055**. Inside a nonconductive electrical junction casing **1059**, the positive insulated conductor **1053** is electrically connected to the positive insulated conductor **916**, and the negative insulated conductor **1055** is electrically connected to the negative insulated conductor **918**. As described above, in the drained polarized plug **913**, the positive insulated conductor **916** is electrically connected to the positive electrical prong **933** and the negative insulated conductor **918** is electrically connected to the negative electrical prong **935**. The drained polarized plug **913** is provided in the source battery connection adapter cable **1050** for electrical connection to the polarized battery clamp **1070**.

(140) The polarized battery clamp **1070** includes a contact handle **1076** with positive battery terminal contact pad **1072** for placement against the positive battery terminal **1062**, and a contact arm **1078** with a negative battery terminal contact pad **1074** for placement against the negative battery terminal **1064**. One end of the contact handle **1076** has a battery clamp electrical receptacle **1080** containing a positive electrical slot **1082** (hidden) and a negative electrical slot **1084** (hidden). The positive electrical slot **1082** is electrically connected to the positive battery terminal contact pad **1072**, and the negative electrical slot **1084** is electrically connected to the negative battery terminal contact pad **1074**. A polarity indicator **1086** illuminates to inform a user whether the polarized battery clamp **1070** has been correctly placed onto the battery terminals **1062**, **1064**. The electrical slots **1082**, **1084** are sized and shaped so as to accommodate insertion of the respective electrical prongs **933**, **935**. To complete a charging setup, the positive electrical prong **1056** is inserted into a positive contact slot **1046** and the negative electrical prong **1058** is inserted into a negative contact slot **1048**. In this charging setup, the driver of the vehicle with the source battery **1040** may provide the source battery connection adapter cable **1050**, and the polarized battery clamp **1070** may be provided by either driver.

(141) FIG. 62 illustrates an exemplary alternative application of the present invention in the process of charging the drained battery **1060**. In this charging configuration, the source battery **880**

is electrically connected to the drained battery **1060** with the battery adapter cable **940** and a booster cable clamp adapter **1090** used at the drained battery **1060**. The booster cable clamp adapter **1090** includes a drained polarized receptacle **1100** that includes a positive electrical contact socket **1102** (hidden) and a negative electrical contact socket **1104** (hidden) for electrically mating to the battery adapter cable **940**. A positive booster cable clamp **1092** is connected to the positive electrical contact socket **1102** by a positive insulated conductor **1096**, and is used for attachment to the positive battery terminal **1062**. A negative booster cable clamp **1094** is connected to the negative electrical contact socket **1104** by a negative insulated conductor **1098**, and is used for attachment to the negative battery terminal **1064**. In this charging setup, the driver of the vehicle with the source battery **880** may provide the battery adapter cable **940**, and the booster cable clamp adapter **1090** may be provided by either driver.

(142) It is to be understood that the description herein is only exemplary of the invention, and is intended to provide an overview for the understanding of the nature and character of the disclosed battery features and configurations, and of the cable connection and adapter assemblies. The accompanying drawings are included to provide a further understanding of various aspects and embodiments of the devices of the invention which, together with their description and claims, serve to explain the principles and operation of the invention.

Claims

1. An automotive battery with a positive battery post providing an electrical path between a positive battery terminal and a positive plate strap, and a negative battery post providing an electrical path between a negative battery terminal and a negative plate strap, said battery comprising: a positive contact slot encasement internal to the battery, said positive contact slot encasement enclosing a positive contact slot and at least a portion of the positive battery post, said positive contact slot having a first opening in a battery housing, said positive contact slot providing access to a positive battery post contact region so as to allow insertion of a positive electrical prong into said first opening whereby a positive prong contact surface is placed into electrical contact with the positive battery post; and, a negative contact slot encasement internal to the battery, said negative contact slot encasement enclosing a negative contact slot and at least a portion of the negative battery post, said negative contact slot having a second opening in said battery housing, said negative contact slot providing access to a negative battery post contact region so as to allow for insertion of a negative electrical prong into said second opening whereby a negative prong contact surface is placed into electrical contact with the negative battery post.
2. The battery of claim 1 wherein said positive battery post contact region comprises a nonplanar rectangle on a surface of the positive battery post, said nonplanar rectangle bounded by a first circumferential arc, a second circumferential arc, a first longitudinal line segment, and a second longitudinal line segment.
3. The battery of claim 1 wherein said first opening comprises an opening in a battery cover.
4. The battery of claim 1 wherein said first opening comprises an opening in a battery casing.
5. The battery of claim 1 wherein said first opening comprises an opening in a recessed surface on an edge of the battery.
6. The battery of claim 1 further comprising a second positive contact slot having a third opening in said battery housing, said second positive contact slot providing access to a second positive battery post contact region so as to allow insertion of a second positive electrical prong into said third opening whereby said second positive electrical prong is placed into electrical contact with the positive battery post.
7. The battery of claim 1 wherein said positive prong contact surface comprises an oblique concave contact edge.
8. An automotive battery with (i) a positive battery terminal on a battery housing connected to a

positive battery post inside the battery housing, the positive battery post providing an electrical path between the positive battery terminal and a positive plate strap in the battery housing, and (ii) a negative battery terminal on the battery housing connected to a negative battery post inside the battery housing, the negative battery post providing an electrical path between the negative battery terminal and a negative plate strap in the battery housing, said battery comprising: a positive contact slot encasement inside the battery housing, said positive contact slot encasement having a positive cavity side wall attached to the battery housing, a positive cavity end wall attached to the battery housing, and a positive cavity base attached to both said positive cavity walls, said positive contact slot encasement enclosing at least a portion of the positive battery post; a positive contact slot having an opening in the battery housing to provide for insertion of a positive electrical prong into said positive contact slot encasement, said positive electrical prong having a contact surface for electrically contacting a positive post electrical contact region on a positive battery post surface when inserted into said positive contact slot encasement; a negative contact slot encasement inside the battery housing, said negative contact slot encasement having a negative cavity side wall attached to the battery housing, a negative cavity end wall attached to the battery housing, and a negative cavity base attached to both said negative cavity walls, said negative contact slot encasement enclosing at least a portion of the negative battery post; and a negative contact slot having an opening in the battery housing to provide for insertion of a negative electrical prong into said negative contact slot encasement, said negative electrical prong having a contact surface for electrically contacting a negative post electrical contact region on a negative battery post surface when inserted into said negative contact slot encasement.

9. The battery of claim 8 wherein said positive battery post contact region comprises a curved band on said positive battery post surface, said curved band bounded by a first circumferential arc, a second circumferential arc, a first longitudinal line segment, and a second longitudinal line segment.

10. The battery of claim 8 wherein said positive cavity base comprises an encasement groove to provide guidance for said positive electrical prong.

11. The battery of claim 8 wherein said positive cavity base comprises a battery post through hole configured to enclose the positive battery post.

12. The battery of claim 8 wherein said positive contact slot comprises a curved encasement channel terminus functioning to force said positive electrical prong against said positive post electrical contact region.

13. The battery of claim 8 wherein said positive cavity end wall comprises a convex surface functioning to force said positive electrical prong against the positive battery post.

14. The battery of claim 8 wherein a longitudinal axis of said positive contact slot forms an acute angle with a longitudinal axis of the positive battery post.

15. A charging system suitable for charging a drained battery using a source battery, said charging system comprising: a source battery connection adapter having a positive source electrical plug for electrically connecting to a source positive battery post via a source positive contact slot, and a negative source electrical plug for electrically connecting to a source negative battery post via a source negative contact slot; a drained battery connection adapter having a positive insulated electrical conductor for electrically connecting to a drained positive battery terminal, and a negative insulated electrical conductor for electrically connecting to a drained negative battery terminal; and, an adapter connection cable configured to be electrically connected both to said source battery connection adapter and to said drained battery connection adapter so as to form a battery interconnection cable assembly such that, when said battery interconnection cable assembly is electrically connected both to the source battery and to the drained battery, said battery interconnection cable assembly provides a positive electrical path between said source positive battery post and said drained positive battery terminal, and provides a negative electrical path between said source negative battery post and said drained negative battery terminal.

16. The charging system of claim 15 wherein said source battery connection adapter comprises a positive electrical socket connected to said positive source electrical plug by a positive insulated conductor, and a negative electrical socket connected to said negative source electrical plug by a negative insulated conductor.

17. The charging system of claim 15 wherein said drained battery connection adapter comprises a polarized battery clamp.

18. The charging system of claim 15 wherein said drained battery connection adapter comprises a booster cable clamp adapter.

19. The charging system of claim 15 wherein said adapter connection cable comprises a first polarized electrical plug at a first end of a cable conductor pair and a second polarized electrical plug at a second end of said cable conductor pair.

20. The charging system of claim 15 further comprising a battery side mount clamp attached to the source battery, said battery side mount clamp housing said source positive contact slot for connecting to said positive source electrical plug, and for housing said source negative contact slot for connecting to said negative source electrical plug.
